VQA as Conditional Domain Transfer via Rectified-Flow (Reflow) JEPA

Abstract

We formalize visual question answering (VQA) as a conditional domain-transfer problem between a visual embedding manifold M_V and a textual embedding manifold M_T, and analyze the resulting decoder through conditional flow matching (CFM) and Rectified Flow (*Reflow*). We show that an L^2-regression decoder can collapse off the (non-convex) manifold M_T on a positive-probability set of prompts (Lemma 1), motivating a generative, flow-based decoder in its place. For the CFM/Reflow construction that follows, we prove, from explicitly stated hypotheses: exact preservation of the target marginal under the learned flow map (Theorem 2); a non-increasing-transport-cost property of Reflow for every convex cost (Theorem 3); a best-iterate $O(1/K)$ rate at which repeated Reflow drives trajectories toward straightness (Theorem 4); and the precise additional structural condition — a convex-potential gradient field — needed for the resulting map to coincide with the quadratic-cost optimal transport (Brenier) map once $d \geq 2$ (Proposition 1). The machinery follows Liu, Gong & Liu (ICLR 2023), adapted here to the conditional VQA setting.

1 Setup

Let $\mathcal{Z} \subset \mathbb{R}^d$ be a shared latent space, $M_V, M_T \subset \mathcal{Z}$ the visual and textual embedding manifolds, $z_v \sim P_V$, $z_t \sim P_T$, and $c \sim P_C$ a prompt conditioning vector. (z_v, z_t) have some joint law given c (e.g. the empirical VQA pairing); independence is *not* assumed anywhere below. All statements are conditional on c; since $\mathbb{E}[\cdot] = \mathbb{E}_c\big[\mathbb{E}[\cdot \mid c]\big]$, any equality or inequality proved for a.e. c integrates through to the unconditional statement, so we suppress "$\mid c$" when unambiguous.

Assumption 1 (Standing integrability)**.** $\mathbb{E}\big[\|z_v\|^2 + \|z_t\|^2 \mid c\big] < \infty$ *a.s. This is used silently throughout: for $\mathcal{L}_{\mathrm{CFM}}$ to be finite, for both Jensen steps in Theorem 3, for the weak continuity-equation formulation in Theorem 2 ($v\, p_\tau \in L^1$), and for the telescoping budget in Theorem 4.*

2 Regression to the Mean off a Non-Convex Manifold

Lemma 1 (Off-manifold collapse of the L^2-optimal predictor)**.** *The minimizer of $\mathcal{L}_{\mathrm{JEPA}}(\theta) = \mathbb{E}\big[\|f_\theta(z_v, c) - z_t\|_2^2\big]$ is $f_\theta^\star(z_v, c) = \mathbb{E}[z_t \mid z_v, c]$, unique up to $P_{V,C}$-null sets (as always for conditional expectations). Moreover, if M_T is non-convex, there exists a data distribution with P_T atomic (finitely many valid answer embeddings — the VQA reality) and a set A of conditioning values (z_v, c) with $P_{V,C}(A) > 0$, such that $f_\theta^\star(z_v, c) \notin M_T$ for every $(z_v, c) \in A$.*

Proof. Minimizer. For fixed (z_v, c), $\mathbb{E}\big[\|f - z_t\|^2 \mid z_v, c\big]$ is a strictly convex quadratic in f; its unique stationary point, found by setting the gradient $2\big(f - \mathbb{E}[z_t \mid z_v, c]\big)$ to zero, is the global minimizer, defined up to $P_{V,C}$-null sets since the objective only integrates against $P_{V,C}$.

Off-manifold existence on a positive-probability set. By definition of non-convexity of M_T, there exist $a, b \in M_T$ and $\lambda \in (0,1)$ with $z^\star := (1-\lambda)a + \lambda b \notin M_T$. Let A be any positive-probability class of prompts admitting exactly the two correct answers $a, b \in M_T$ (well-defined once P_T is atomic, since then a positive-probability set of (z_v, c) genuinely shares the same two-point conditional law) and set $P_T(\cdot \mid z_v, c) = (1-\lambda)\delta_a + \lambda\delta_b$ for $(z_v, c) \in A$. Then for every $(z_v, c) \in A$, $f_\theta^\star(z_v, c) = (1-\lambda)a + \lambda b = z^\star \notin M_T$ — a failure on a set of positive probability, not a $P_{V,C}$-null artifact that the loss never sees. □

Remark 1. The only fact used is the defining property of non-convexity itself; no appeal to Jensen's inequality is needed or correct here (Jensen compares $\varphi(\mathbb{E}X)$ to $\mathbb{E}[\varphi(X)]$ for convex φ, which is not the statement being made). Not every multimodal law on a non-convex M_T collapses off-manifold — symmetric cases can coincide with a point of M_T by chance — so the Lemma is an existence result on a non-null event, not a universal claim: it shows the failure mode is real, empirically visible, and unavoidable in general, which is what is needed to motivate Section 3. It does *not*, by itself, force retrieval-based decoding as the unique fix — CFM below is a constructive counterexample to that stronger claim.

3 Conditional Flow Matching

Define $z_\tau = (1-\tau)z_v + \tau z_t$ and the CFM objective

$$\mathcal{L}_{\mathrm{CFM}}(\theta_1) = \mathbb{E}_{\tau, z_v, z_t, c}\Big[\big\|v_{\theta_1}(z_\tau, \tau, c) - (z_t - z_v)\big\|_2^2\Big], \qquad \tau \sim \mathrm{Unif}[0,1].$$

Assumption 2 (Realizability)**.** *v_{θ_1} attains the exact population minimizer of $\mathcal{L}_{\mathrm{CFM}}$, i.e.*

$$v_{\theta_1}(z, \tau, c) = \mathbb{E}[z_t - z_v \mid z_\tau = z,\, c] \qquad a.e.$$

(unique up to null sets, by the same stationarity argument as Lemma 1).

Remark 2 (This assumes away, rather than resolves, the singularity of §4)*.* The conditional-mean field $\mathbb{E}[z_t - z_v \mid z_\tau = z, c]$ can fail to be Lipschitz precisely where straight-line paths of the underlying data cross (§4). Assumption 2 together with Assumption 3 require the *realized* v_{θ_1} to already be Lipschitz — i.e., to have already resolved any such singularity, the same restriction Liu, Gong & Liu make in confining their analysis to "rectifiable" couplings. The theorems below describe the idealized, well-posed iteration once round 0 is well-posed; they do not show that an arbitrary, possibly ill-posed empirical coupling yields a well-posed v_{θ_1} in the first place — that is assumed, not derived.

4 Trajectory Crossing and Well-Posedness

Theorem 1 (Picard–Lindelöf, sufficient condition)**.** *If, for each c, $v_{\theta_1}(\cdot, \cdot, c)$ is continuous in $\tau \in [0,1]$, Lipschitz in z uniformly in τ, and satisfies a linear growth bound $\|v_{\theta_1}(z, \tau, c)\| \le A(c)(1 + \|z\|)$, then the IVP $\dot{z}_\tau = v_{\theta_1}(z_\tau, \tau, c)$, $z_0 = z_v$, has a unique solution on all of $[0,1]$.*

This is a *sufficient*, not necessary, condition for uniqueness; we invoke it as such and do not claim the converse.

Assumption 3 (Well-posedness)**.** *v_{θ_1} satisfies the hypotheses of Theorem 1, so the time-1 flow map $\Phi_1(\cdot, c) : \mathcal{Z} \to \mathcal{Z}$, $\Phi_1(z_v, c) := z_v + \int_0^1 v_{\theta_1}(Z_\tau, \tau, c)\, d\tau$ (where Z_τ solves the IVP), is well-defined.*

The singularity problem. If two straight lines $z_v^{(1)} \to z_t^{(1)}$ and $z_v^{(2)} \to z_t^{(2)}$ cross at (z', τ'), the pointwise conditional-mean target $\mathbb{E}[z_t - z_v \mid z_{\tau'} = z', c]$ need not vary continuously with z' near the crossing — it is a well-defined single number at each point (an expectation), but can fail to be Lipschitz there, which is precisely the regularity Theorem 1 requires. This motivates rectification below; it is a motivation, not a proof that Reflow's output is exactly straight after one round — see Theorem 4, which shows this happens only in a best-iterate, $O(1/K)$ sense over repeated rounds.

5 Rigorous Properties of Reflow

Let $\hat{z}_t := \Phi_1(z_v, c)$ and Z_τ the corresponding flow path, under Assumptions 2–3.

Theorem 2 (Marginal Preservation)**.** *For every* $\tau \in [0,1]$, $\mathrm{Law}(Z_\tau \mid c) = \mathrm{Law}(z_\tau \mid c)$. *In particular, at* $\tau = 1$*:* $\hat{z}_t \mid c \;\sim\; P_T(\cdot \mid c)$ *exactly, so* $\hat{z}_t \in \operatorname{supp} P_T(\cdot \mid c)$ *almost surely; if* M_T *is closed — e.g.* $M_T := \operatorname{supp} P_T(\cdot \mid c)$, *automatic when* P_T *is atomic as in VQA's discrete answer space — this reads* $\hat{z}_t \in M_T$ *a.s.*

Proof. For any compactly supported C^1 test function h,

$$\frac{d}{d\tau}\mathbb{E}[h(z_\tau) \mid c] = \mathbb{E}\big[\nabla h(z_\tau)^\top (z_t - z_v) \mid c\big] = \mathbb{E}\Big[\nabla h(z_\tau)^\top\, \mathbb{E}[z_t - z_v \mid z_\tau, c] \mid c\Big] = \mathbb{E}\big[\nabla h(z_\tau)^\top v_{\theta_1}(z_\tau, \tau, c) \mid c\big],$$

using the tower property and Assumption 2 (Assumption 1 makes each term finite). Hence $p_\tau := \mathrm{Law}(z_\tau \mid c)$ solves, weakly, the continuity equation $\partial_\tau p_\tau + \nabla \cdot (v_{\theta_1}(\cdot, \tau, c)\, p_\tau) = 0$, $p_0 = P_V(\cdot \mid c)$. The pushforward of $P_V(\cdot \mid c)$ under the flow of $v_{\theta_1}(\cdot, \tau, c)$ — i.e. $\mathrm{Law}(Z_\tau \mid c)$ — solves the identical linear PDE with the identical initial condition (the Liouville/continuity equation associated to any well-posed ODE flow). Under Assumption 3, the solution to this continuity equation is unique (Ambrosio & Crippa, 2008; Kurtz, 2011), so $\mathrm{Law}(Z_\tau \mid c) = p_\tau$ for all τ, and at $\tau = 1$, $p_1 = P_T(\cdot \mid c)$ by construction of z_τ. □

Remark 3. The equality just proved is *marginal*: $\mathrm{Law}(Z_\tau \mid c) = \mathrm{Law}(z_\tau \mid c)$ at each fixed τ, not pathwise. Z is the causalized, derandomized version of the (in general non-Markov, anticipating) interpolation process z_τ; the two processes have different joint laws over $\tau \in [0,1]$. Conclusion 1 of §6 should be read at this marginal level only.

Theorem 3 (Reflow Does Not Increase Convex Transport Cost)**.** *For any convex* $\gamma : \mathbb{R}^d \to \mathbb{R}$ *with* $\mathbb{E}[\gamma(z_t - z_v)] < \infty$*,*

$$\mathbb{E}\big[\gamma(\hat{z}_t - z_v)\big] \;\leq\; \mathbb{E}\big[\gamma(z_t - z_v)\big].$$

Proof. Writing $\hat{z}_t - z_v = \int_0^1 v_{\theta_1}(Z_\tau, \tau, c)\, d\tau$ and using $\tau \sim \mathrm{Unif}[0,1]$ to view the integral as an average:

$$\mathbb{E}\big[\gamma(\hat{z}_t - z_v)\big] \overset{\text{Jensen}}{\leq} \mathbb{E}\Big[\int_0^1 \gamma\big(v_{\theta_1}(Z_\tau, \tau, c)\big)\, d\tau\Big] = \int_0^1 \mathbb{E}\big[\gamma(v_{\theta_1}(Z_\tau, \tau, c))\big]\, d\tau.$$

By Theorem 2, $Z_\tau \overset{d}{=} z_\tau$ at each fixed τ, and $\gamma(v_{\theta_1}(\cdot, \tau, c))$ is a fixed measurable function of the state, so

$$= \int_0^1 \mathbb{E}\big[\gamma(v_{\theta_1}(z_\tau, \tau, c))\big]\, d\tau = \mathbb{E}\Big[\int_0^1 \gamma\big(\mathbb{E}[z_t - z_v \mid z_\tau, c]\big)\, d\tau\Big] \overset{\text{Jensen}}{\leq} \mathbb{E}\Big[\int_0^1 \mathbb{E}[\gamma(z_t - z_v) \mid z_\tau, c]\, d\tau\Big] = \mathbb{E}[\gamma(z_t - z_v)].$$

(Jensen's inequality is applied twice above, as conditional Jensen.) □

Theorem 4 (Recursive Straightening at a Best-Iterate Rate $O(1/K)$)**.** *Let $Z^{(0)} := (z_v, z_t)$, and for $k \geq 0$ let $Z^{(k+1)}$ be the rectified flow trained by CFM (Assumptions 2–3, imposed inductively at every round k) on the coupling $\big(Z_0^{(k)}, Z_1^{(k)}\big)$ produced by simulating $Z^{(k)}$. Define the straightness $S(Z^{(k+1)}) = \int_0^1 \mathbb{E}\big[\|(Z_1^{(k+1)} - Z_0^{(k+1)}) - v_{\theta_{k+1}}(Z_\tau^{(k+1)}, \tau, c)\|^2\big] d\tau$ and crossing measure $V\big(Z_0^{(k)}, Z_1^{(k)}\big) = \int_0^1 \mathbb{E}\big[\|(Z_1^{(k)} - Z_0^{(k)}) - \mathbb{E}[Z_1^{(k)} - Z_0^{(k)} \mid W_\tau^{(k)}]\|^2\big] d\tau$, where $W_\tau^{(k)} = (1-\tau)Z_0^{(k)} + \tau Z_1^{(k)}$. Then*

$$\min_{0 \leq k < K} \Big[S(Z^{(k+1)}) + V\big(Z_0^{(k)}, Z_1^{(k)}\big)\Big] \;\leq\; \frac{\mathbb{E}[\|z_t - z_v\|^2]}{K} = O(1/K).$$

Proof. Take $\gamma(x) = \|x\|^2$ in Theorem 3. For squared norm, both Jensen steps in that proof are equalities up to an exact remainder, via the variance identity

$$\mathbb{E}\|X\|^2 - \mathbb{E}\big\|\mathbb{E}[X \mid \mathcal{F}]\big\|^2 = \mathbb{E}\big\|X - \mathbb{E}[X \mid \mathcal{F}]\big\|^2$$

applied to each conditioning step. Unwinding both remainders gives the exact identity

$$\mathbb{E}\big[\|Z_1^{(k)} - Z_0^{(k)}\|^2\big] - \mathbb{E}\big[\|Z_1^{(k+1)} - Z_0^{(k+1)}\|^2\big] = S(Z^{(k+1)}) + V\big(Z_0^{(k)}, Z_1^{(k)}\big).$$

Both terms on the right are non-negative. Summing (telescoping) over $k = 0, \ldots, K-1$:

$$\sum_{k=0}^{K-1} \Big[S(Z^{(k+1)}) + V\big(Z_0^{(k)}, Z_1^{(k)}\big)\Big] = \mathbb{E}[\|z_t - z_v\|^2] - \mathbb{E}\big[\|Z_1^{(K)} - Z_0^{(K)}\|^2\big] \;\leq\; \mathbb{E}[\|z_t - z_v\|^2].$$

The minimum of K non-negative summands is at most their average, giving the stated bound. □

Remark 4. This is a *best-iterate* guarantee: it exhibits some round $k^\star < K$ with $S + V \leq \mathbb{E}\|z_t - z_v\|^2/K$ (equivalently, $\liminf_k \big[S(Z^{(k+1)}) + V(Z_0^{(k)}, Z_1^{(k)})\big] = 0$). It does *not* assert that the sequence $S_k + V_k$ is monotone, that it converges, or that any specific finite round achieves $V = 0$ exactly — only that rounds with small $S + V$ exist, and become easier to find as K grows. In particular, non-crossing after a *single* Reflow step is not guaranteed.

Proposition 1 (Straightness Is Necessary but Not Sufficient for Exact OT, $d \geq 2$)**.** *A coupling with $V = 0$ (a fixed point of Reflow) need not be optimal for a given strictly convex cost γ once $d \geq 2$: a pure rotation of $\mathcal{Z}$ is Lipschitz (so Theorem 1 applies) and induces a deterministic, non-crossing coupling, yet is not the OT map for a rotation-invariant P_T (the identity map has strictly lower quadratic cost). Matching the Brenier/Monge optimal transport map for the quadratic cost requires the additional, non-automatic restriction that the learned field be a gradient field, $v_\theta(\cdot, \tau, c) = \nabla_z f_\tau(\cdot, c)$ for some scalar potential f_τ — and, for exact identification with the quadratic-cost Brenier map specifically, that $f_\tau(\cdot, c)$ be* convex*: a gradient field with a non-convex potential need not be optimal either. The gradient-field restriction removes exactly the rotational component that an unconstrained least-squares fit is free to retain.*

Remark 5. The rotation's straight-line chords *do* cross as geometric segments in $\mathcal{Z}$, but never at the *same* τ: V measures simultaneous-time crossing — whether position at time τ determines velocity, which is what the vector field must resolve — and crossing at different times is invisible to V. This is not a technicality specific to the VQA setting: it is the published content of Liu, Gong & Liu (2023, §3.4) and Liu (2022, "On Rectified Flow and Optimal Coupling"). Vanilla Reflow provably decreases cost and straightens (Theorems 3–4); it provably does *not*, on its own, converge to a specific γ-optimal coupling in $d \geq 2$ without the gradient-field restriction above. The one-dimensional case is the exception where straightness and γ-optimality for every convex γ coincide.

6 Conclusion

Under Assumptions 1–3 (finite second moments; exact population-level fit; well-posed ODE):

1. **Manifold adherence.** By Theorem 2, $\hat{z}_t \in M_T$ *exactly* (at the marginal level, not pathwise) — but this is a statement about the idealized population-optimal v_{θ_1} with exact integration, not a guarantee that survives finite data, finite network capacity, or discretized Euler integration without error.

2. **Inference efficiency.** If the k-th rectified flow is fully straight, $S(Z^{(k)}) = 0$, then a single Euler step $\hat{z}_t = z_v + v_{\theta_k}(z_v, 0, c)$ is exact (constant velocity along the trajectory gives $v_{\theta_k}(Z_0, 0, c) = Z_1 - Z_0$). The converse is *not* claimed: one-step exactness from $\tau = 0$ alone only pins down the field at $\tau = 0$ and does not by itself imply $S(Z^{(k)}) = 0$; the genuine equivalence is between $S(Z^{(k)}) = 0$ and Euler-step exactness from *every* point (τ, Z_τ) along the trajectory, i.e. $v_{\theta_k}(Z_\tau, \tau, c) = (Z_1 - Z_\tau)/(1 - \tau)$ for all τ. Theorem 4 shows a round with small $S + V$ exists within K rounds of Reflow at a best-iterate rate $O(1/K)$ — it does not guarantee $S = 0$ after a single round. Per Proposition 1, exact alignment with a specific optimal-transport plan additionally requires the gradient-field (and, for the quadratic cost, convex-potential) restriction. In practice, near-straight flows from a modest number of Reflow rounds are typically finished off with a distillation step to get accurate one-step sampling; this is an engineering addition, not a free consequence of Reflow alone.

7 Methodology: Training-Procedure Substitution for Controlled Comparison

The theory of §§2–5 makes falsifiable predictions about a concrete system. To test them without confounding the result with an architecture change, we hold the VL-JEPA backbone fixed and substitute only the training procedure and the inference-time decoding rule.

7.1 Shared Architecture

All arms share: a visual encoder E_V producing $z_v = E_V(\text{image}, \text{question}) \in \mathcal{Z}$; a text encoder E_T that embeds the (finite, VQA-atomic) answer vocabulary $\mathcal{A}$ into $M_T := E_T(\mathcal{A}) \subset \mathcal{Z}$, giving $z_t = E_T(\text{answer})$ at training time; and VL-JEPA's predictor network, at identical depth, width, and conditioning pathway for the prompt vector c. The only architectural change is a time-conditioning input: the predictor $f_\theta(z_v, c)$ is repurposed as $v_\theta(z_\tau, \tau, c)$ by injecting a sinusoidal embedding of $\tau \in [0, 1]$ through the same mechanism VL-JEPA already uses to inject c (e.g. FiLM or cross-attention), adding one embedding table and its injection layer and nothing else. Encoder weights, predictor depth/width, and the conditioning pathway for c are identical to the baseline. For the image-QA phase below, E_V operates on a single image rather than a video-frame sequence; this removes VL-JEPA's temporal pooling/frame-sampling machinery as a degree of freedom, so the comparison is only ever about the training procedure, never about how frames were sampled or pooled.

7.2 Arm (A): Baseline VL-JEPA Training

Trained exactly as VL-JEPA specifies: minimize $\mathcal{L}_{\text{JEPA}}(\theta) = \mathbb{E}[\|f_\theta(z_v, c) - z_t\|_2^2]$ (Lemma 1). At inference, $f_\theta(z_v, c)$ is scored by InfoNCE similarity against the fixed candidate embeddings M_T and decoded by top-1 (or top-k) retrieval; the network never outputs a token sequence directly.

7.3 Arms (B)–(D): CFM + Reflow Training

Given the same (z_v, z_t, c) triples used to train arm (A), $\mathcal{L}_{\text{JEPA}}$ is replaced by the finite-sample CFM objective of §3, iterated via Reflow (Algorithm 1). Round $k = 1$ (arm B) trains directly on the data coupling; rounds $k = 2, \ldots, K$ (arm C, for K chosen by the sweep below) retrain on the deterministic coupling produced by simulating the previous round's field, matching the construction analyzed in Theorem 4. Arm (D) adds the optional one-step distillation of Algorithm 2 on top of arm (C)'s final field.

Algorithm 1 CFM + Reflow training for the predictor v_θ

Require: data $\mathcal{D} = \{(z_v^{(i)}, z_t^{(i)}, c^{(i)})\}_{i=1}^n$; number of Reflow rounds K; Euler steps per simulated trajectory N; VL-JEPA predictor f_θ augmented with the time input above

Initialize θ_1 from the pretrained VL-JEPA backbone (time-embedding parameters random)
$\mathcal{D}_1 \leftarrow \mathcal{D}$
for $k = 1$ **to** K **do**
  Train v_{θ_k} to convergence on $\mathcal{D}_k$ by minimizing $\widehat{\mathbb{E}}_{\tau, z_v, z_t, c}\big[\|v_{\theta_k}(z_\tau, \tau, c) - (z_t - z_v)\|_2^2\big]$, $\tau \sim \text{Unif}[0, 1]$
  Log $\widehat{\mathbb{E}}\|z_t - z_v\|^2$, the straightness/crossing proxy $\hat{S} + \hat{V}$, and the manifold-adherence rate on a held-out split
  if $k < K$ **then**
    for each $(z_v^{(i)}, c^{(i)}) \in \mathcal{D}$ **do**
      $\hat{z}_t^{(i)} \leftarrow N$-step Euler integration of v_{θ_k} from $z_v^{(i)}$
    end for
    $\mathcal{D}_{k+1} \leftarrow \{(z_v^{(i)}, \hat{z}_t^{(i)}, c^{(i)})\}_{i=1}^n$
    $\theta_{k+1} \leftarrow \theta_k$ (warm start)
  end if
end for
return v_{θ_K}

Algorithm 2 Optional one-step distillation (arm D only)

Require: trained field v_{θ_K}; teacher Euler steps N

for each $(z_v^{(i)}, c^{(i)}) \in \mathcal{D}$ **do**
  $\hat{z}_t^{(i)} \leftarrow N$-step Euler integration of v_{θ_K} from $z_v^{(i)}$
end for
Train a student head $g_\phi(z_v, c)$ — same backbone, no time input — by minimizing $\mathbb{E}\|g_\phi(z_v, c) - \hat{z}_t\|^2$
return g_ϕ

7.4 Inference and Decoding

Arms (B) and (C) decode by M-step Euler integration of the trained field from z_v ($M = N$ for (B); $M \in \{1, 2, 4, 8\}$ swept for (C) to trace the accuracy-vs-NFE curve implied by the Conclusion's inference-efficiency discussion), followed by a nearest-neighbor snap of the resulting $\hat{z}_t$ onto M_T — exact-manifold decoding rather than arm (A)'s InfoNCE retrieval, made possible by Theorem 2. Arm (D) decodes with a single forward pass, $\hat{z}_t = g_\phi(z_v, c)$, snapped the same way. All four arms use the identical snap-to-M_T (or, for (A), identical InfoNCE) rule and the identical text realization of the nearest embedding, so decoded-text differences are attributable only to the predicted embedding.

7.5 Evaluation Protocol

We evaluate in two stages. **Phase 1 (image QA)** isolates the training-procedure effect on the simplest instance of the problem, with no temporal machinery and comparatively cheap Reflow rounds; **Phase 2 (video QA)** is undertaken only once Phase 1 clears the convergence criterion of §7.6. Phase 1 uses GQA — compositional visual reasoning, one of the four VQA benchmarks VL-JEPA itself reports — and VQAv2, whose ten independent human answers per question directly instantiate the atomic, multi-answer P_T construction of Lemma 1; POPE's binary yes/no answer space ($|M_T| = 2$) additionally serves as a minimal-scale correctness check on the CFM/Reflow implementation before the full sweep. All arms are trained and evaluated on the identical splits of each benchmark, so the training procedure is the only varying factor relative to arm (A). We report:

1. **Answer accuracy** — standard VQA accuracy on the decoded text, computed identically across arms.
2. **Manifold-adherence rate** — $\Pr[\mathrm{dist}(\hat{z}_t, M_T) \le \varepsilon]$ on held-out data, *before* the nearest-neighbor snap, for a small fixed ε; the empirical counterpart of Theorem 2, expected to separate (A) from (B)–(D).
3. **Multi-answer coverage** — recall@k against the full set of human-annotated acceptable answers, restricted to questions with ≥ 2 accepted answers; the case Lemma 1 targets, again expected to separate (A) from (B)–(D).
4. **Inference cost** — NFE and wall-clock latency at matched accuracy; expected to separate (B) from (C)/(D), testing the Conclusion's efficiency claim.
5. **Diagnostic curves vs. round** k — $\widehat{\mathbb{E}}\|\hat{z}_t - z_v\|^2$ and $\hat{S} + \hat{V}$ from Algorithm 1's logging, reported for arm (C) across $K \in \{1, 2, 4, 8\}$, to check the non-increasing transport cost of Theorem 3 and the $O(1/K)$ trend of Theorem 4 directly against data.

Arm	Training objective	Decoding	Isolates
(A) VL-JEPA	$\mathcal{L}_{\mathrm{JEPA}}$	InfoNCE retrieval over $\mathcal{A}$	—
(B) + CFM	$\mathcal{L}_{\mathrm{CFM}}$, $K = 1$, N-step Euler	snap to M_T	Lemma 1, Thm 2
(C) + Reflow	$\mathcal{L}_{\mathrm{CFM}}$, K rounds, M-step Euler	snap to M_T	Thm 3, Thm 4
(D) + distillation	(C) + Algorithm 2	$g_\phi(z_v, c)$, single pass	Conclusion, pt. 2

Table 1: Ablation ladder. Encoders, the candidate/answer set $\mathcal{A}$, and the predictor backbone are identical across arms; only the training procedure and the inference-time integration rule differ.

7.6 Convergence Criterion and Staging to Video

For arm (C), sweep $K \in \{1, 2, 3, 4, 6, 8\}$ on GQA and VQAv2 and, for each K, plot three quantities against the round index: the validation transport cost $\widehat{\mathbb{E}}\|\hat{z}_t - z_v\|^2$, the diagnostic $\hat{S} + \hat{V}$, and downstream VQA accuracy. Theorems 3–4 predict the first two are non-increasing and $O(1/K)$ respectively; the empirical claim under test is that accuracy tracks them — rising and flattening rather than continuing to climb — which would indicate the predicted representation-quality trend is what drives any accuracy gain, rather than an unrelated effect of extra training compute.

We pre-register the Phase 1 success condition before running Phase 2: arm (C), at the K where the diagnostic curve flattens, must exceed arm (A) on accuracy and on the manifold-adherence rate on *both* GQA and VQAv2, and must exceed arm (B)'s NFE-matched accuracy (testing that Reflow, not CFM alone, is doing the work). Only if this holds do we proceed to Phase 2, extending E_V to VL-JEPA's video-frame input and evaluating on its own video classification, video retrieval, and world-modeling benchmarks — at which point Reflow's per-round compute cost (§7) is the dominant practical constraint, making the K found in Phase 1 the natural starting point rather than a re-swept parameter.

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